Cultural Matching and the Persistence of Social Ties

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How do cultural tastes shape the evolutionary dynamics of personal networks? While traditional perspectives often view tastes as malleable byproducts of network transmission, we advance a cultural matching model where pre-existing tastes actively determine tie persistence. Using longitudinal ego-network data from the NetSense study, we find strong evidence that shared cultural tastes—both broad categorical interests and specific shared activities—are vital mechanisms for stabilizing social bonds. Conversely, cultural network opacity, or a lack of mutual cultural knowledge, acts as a powerful liability that accelerates tie decay. Crucially, these effects are heavily concentrated among strong ties: cultural alignment forms the essential structural core for maintaining subjectively close relationships, whereas weak ties face such a high baseline hazard of decay that cultural matching alone cannot save them. Ultimately, these findings suggest that cultural tastes function as durable resources that individuals actively use to maintain their most important social connections.

1 Introduction

A central issue in the study of culture and networks concerns the co-evolution of social ties and cultural tastes (Lizardo, 2006; Edelmann and Vaisey, 2014). Traditional sociological models of taste formation posit that an individual's cultural profile is largely shaped by their current social network configuration (Mark, 1998b; McPherson, 2004). Tastes are assumed to diffuse through existing homophilous ties, which are themselves structured by socio-demographic similarity—"status homophily" (Lazarsfeld and Merton, 1954; McPherson et al., 2001; Feld and Grofman, 2009). Under this view, "birds of a feather flock together," and, as a result, they end up consuming the same cultural goods and developing similar tastes (Mark, 2003). Tastes are construed as highly malleable, "thin" contents liable to shift whenever the surrounding network ties change. Yet longitudinal network research reveals that personal networks are in constant flux, with ties being added, modified, and deleted at presumably faster time-scales than tastes (Burt, 2000, 2002; Wellman et al., 1997; Kleinbaum, 2018; Shipilov et al., 2006). This empirical fact of network volatility raises a major question: if networks are unstable and tastes are merely thin, network-contingent contents, how do we account for the durable, resilient cultural profiles that individuals maintain over time (Bourdieu, 1984; Holt, 1998)?

In this paper, we provide empirical evidence consistent with a **cultural matching model** of tie persistence. We propose that, rather than tastes merely adapting to match pre-existing networks, pre-existing cultural dispositions serve as an active "social glue" that determines which network ties survive and which dissolve (Burt, 2000). Under this model, dyadic cultural matching (commonalities in tastes and activities) increases the over-time stability and cognitive salience of social ties, net of usual dyadic determinants such as subjective closeness and tie duration. Using rich longitudinal ego-network data from the *NetSense* study over eight waves (Bahulkar et al., 2016, 2017; Hachen et al., 2022), we track the persistence of social ties across multiple college-year transitions. To isolate peer dynamics, we focus on ties among non-kin, on-campus students. Empirically, we unpack the cultural matching mechanism in two ways. First, we establish the baseline protective effect of aggregate cultural matching on tie persistence. Second, we explore the role of "cultural network opacity"—the degree to which an ego lacks knowledge of their alter's preferences (indicated by "Don't Know" responses)—as an independent predictor of tie decay.

The remainder of this article proceeds as follows. First, we outline the theoretical framework linking tastes, social networks, and tie persistence. Next, we describe the NetSense data and our person-period modeling strategy. Finally, we present the results and discuss their implications for the study of homophily and network evolution.

2 Theoretical Framework

2.1 Tastes and Social Networks: Two Perspectives

The relationship between cultural tastes and social networks has been conceptualized in two contrasting ways in sociological theory. The **network transmission perspective**, drawing on constructural imagery (Carley, 1991), views tastes as malleable, network-contingent contents

(Erickson, 1988, 1996; Mark, 1998a, 2003). In this view, social ties constitute the pre-existing "infrastructure" or "pipes" through which cultural practices diffuse (Borgatti, 2005; Erickson, 1996). Tastes are learned, adopted, and abandoned based on continuous exchange of cultural information and local bandwagon imitation within an actor's immediate ego network (Mark, 2003). Because personal networks exhibit substantial over-time volatility—with 30% to 60% turnover annually in most personal networks (Morgan et al., 1997; Suitor et al., 1997)—the network transmission perspective implies that individual tastes must exhibit corresponding volatility.

In contrast, the *cultural capital perspective* conceptualizes tastes as highly durable, embodied dispositions reaped from early socialization (Bourdieu, 1984, 1990; Holt, 1997, 1998). Emphasizing the "interconvertibility" of different forms of capital (Bourdieu, 1986), this view sees cultural aptitudes not merely as by-products of ties, but as fundamental resources that help people form and sustain network connections. Here, networks are seen as fleeting structures built and maintained via interactions (DiMaggio, 1987). Consequently, individuals actively use pre-existing tastes to construct novel relationships or shed culturally incompatible ties, a dynamic formalized in the cultural matching model (Lizardo, 2006; Vaisey and Lizardo, 2010). In this framework, internalized cultural practices exert an independent causal effect in shaping social networks because people self-select into ties based on the match between their tastes and others' (Shalizi and Thomas, 2011).

Building on this insight, recent theoretical elaborations specify how matching and influence operate simultaneously as dual mechanisms. For instance, Lewis and Kaufman (2018) proposes a generalized conversion model demonstrating that cultural conversion (tastes diffusing through ties) and cultural matching (ties forming based on shared tastes) operate in tandem, constrained by the local cultural ecology. Crucially for our present study, the degree of cultural match at the dyadic level fundamentally determines whether relationships endure over time or instead selectively decay (Martin and Yeung, 2006). If tastes are durable and personal networks are volatile, it is highly likely that tastes serve as *drivers* of network connections rather than being determined exclusively by them. This leads to the concept of **choice homophily** or cultural matching: new social connections are selected, and existing connections are maintained, based on pre-existing cultural compatibility (Lazarsfeld and Merton, 1954; Werner and Parmelee, 1979). In other words, socio-demographic homophily may frequently operate as a *by-product* of active contact selection keyed around cultural alignment, rather than homophily acting as an exogenous structural constraint (Carley, 1991; DiMaggio, 1987).

2.2 Cultural Matching and Tie Persistence

A major challenge for any newly formed social relationship is what Burt (2000), drawing on Stinchcombe (1965), refers to as the **"liability of newness."** Newly formed dyadic ties are fragile and exhibit a high baseline hazard of decay ("tie decay"), which steeply declines as a relationship matures and ripens into close friendship. What stabilizes a tie and helps it overcome this liability of newness? While classic network models emphasize structural factors (such as network constraint or multiplexity) and dyadic factors (such as subjective closeness), we propose that **cultural matching** is a vital stabilizing mechanism. Rather than just indicating demographic similarity, shared tastes and cultural practices act as an active "social glue." At the micro-level, this matching stabilizes ties by reducing interactional friction, generating positive emotional energy during shared rituals (Collins, 2004), and providing a common cognitive and behavioral baseline that makes the tie more salient and rewarding to maintain (DiMaggio, 1987).

When ego and alter share cultural dispositions, they can engage in shared activities and coordinate behavior with minimal friction. Over time, this cultural alignment increases the cognitive salience of the alter, rendering the tie resistant to decay and facilitating its retention in the ego's active network (Burt, 2000). Therefore, we expect that commonalities in cultural tastes and leisure practices will significantly reduce the hazard of tie decay, net of subjective closeness and structural controls (**The Baseline Matching Hypothesis**).

2.3 Cultural Network Opacity

If shared cultural knowledge stabilizes ties, then a *lack* of mutual cultural knowledge—*cultural network opacity*—should act as a major, independent mechanism of tie decay. We define cultural network opacity as the degree to which an ego is ignorant of an alter's cultural preferences. It is theoretically crucial to differentiate opacity from mere "taste dissimilarity." A taste mismatch implies that ego and alter know each other's preferences but do not share them. Opacity, in contrast, implies a relational void. When an ego reports "Don't Know" regarding an alter's preferences across multiple domains, it signals a profound lack of interactional investment. Furthermore, it creates a severe cognitive barrier to finding focal points for shared activities. Without a map of the alter's cultural landscape, the ego cannot easily initiate shared rituals or conversations, independently predicting a higher hazard of tie dissolution (Burt, 2002; Wellman et al., 1997). We test both the main effect of cultural network opacity on tie decay and the moderation of the protective effect of known cultural matches by opacity.

2.4 Tie Strength and the Limits of Cultural Matching

Furthermore, recent research highlights a critical theoretical tension regarding how the effects of cultural matching intersect with tie strength. On the one hand, drawing on the work of Edelmann and Vaisey (2014) and Vaisey and Lizardo (2010), one might expect that cultural matching is primarily associated with the maintenance of *strong* ties. In this view, deep, institutionalized friendships require ongoing cultural resonance and shared practices to sustain intimacy over time, making close ties highly sensitive to cultural matching (**The Strong-Tie Matching Hypothesis**).

On the other hand, a competing perspective drawn from structural network theory suggests that strong ties are highly resilient due to emotional inertia, multiplexity, and structural embeddedness, rendering them less dependent on specific cultural similarities to survive. Under this view, cultural matching acts as essential *scaffolding* specifically for *weak* or nascent ties. Weak ties suffer from a high baseline liability of newness (Burt, 2000); without shared interests to grease the wheels of interaction and provide focal points for engagement, they rapidly decay. Thus, the protective power of matching should be most pronounced for weak ties (**The**

Weak-Tie Scaffolding Hypothesis). We estimate interaction models to adjudicate between these competing predictions.

2.5 Summary of Hypotheses

In sum, we test three core hypotheses regarding the role of culture in network evolution:

1. **Baseline Matching Hypothesis**: Shared cultural tastes and activities significantly reduce the hazard of tie decay.
2. **Cultural Network Opacity Hypothesis**: A lack of mutual cultural knowledge independently increases the hazard of tie decay.
3. **Moderation by Tie Strength**: We test two competing predictions regarding moderation:
   - **H3a (Strong-Tie Matching)**: The protective effect of cultural matching is amplified among subjectively close ties.
   - **H3b (Weak-Tie Scaffolding)**: The protective effect of cultural matching is amplified among weak ties to overcome the liability of newness.

3 Data and Methods

Data for this analysis comes from the *NetSense* study, a longitudinal survey of college students conducted at the University of Notre Dame between 2011 and 2015. The study collected extensive panel data on students' social networks, demographic backgrounds, and cultural preferences.[1] We collected information on the cultural habits of ego-alter pairs using both a series of closed-form domain questions and open-ended activity nominations. For the closed-form items, egos were asked to rate both their own and their alters' level of interest across six broad cultural and leisure domains: music, movies, books, sports, games, and outdoor activities. For the open-ended items, respondents freely nominated up to five leisure activities they enjoyed, and then indicated whether their alter also engaged in each nominated activity.

3.1 Study Waves

The NetSense study is a longitudinal survey of college students collected over multiple waves (Striegel et al., 2013). For this analysis of tie persistence, we pool data across adjacent waves to construct a person-period dataset. To isolate peer dynamics, we restrict our sample to non-kin, on-campus ties (i.e., alters who are also students at Notre Dame). To capture relationship dynamics across the entirety of the college experience, we construct a discrete-time person-period dataset spanning all available adjacent survey waves (Wave 1 through Wave 8). This design allows us to observe tie persistence continuously across varying intervals of the

[1] A comprehensive codebook, data dictionary, and repository for the NetSense project are available at https://olizardo.github.io/NetSense-Codebook/.

students' academic careers, capturing both the initial network churn of freshman year and the consolidation of networks approaching graduation.

3.2 Measures

3.2.1 Outcome Variable

Tie Persistence: A binary indicator taking a value of 1 if a social tie reported by an ego in a baseline wave (e.g., Wave 1) is still present in the subsequent wave (e.g., Wave 2), and 0 if the tie has dissolved (i.e., the alter is no longer listed in the ego's network). To determine tie presence, egos were asked to list the people with whom they interact most frequently, with the prompt: "Please list the people with whom you have spent the most time in the last month." Egos could nominate up to 20 alters in each wave.

3.2.2 Main Predictors of Interest

- *Closed-Form Cultural Matching*: A count variable (ranging from 0 to 6) indicating the number of closed-form cultural domains in which the ego and alter share similar tastes. Egos were asked to rate their own interest, and their alters' interest across six broad domains: music, movies, books, sports, games, and outdoor activities. The specific survey prompt for egos was: "How interested are you in the following activities?" and for alters: "How interested is this person in the following activities?" Response categories included "Very interested," "Somewhat interested," and "Not at all interested." A domain was scored as a "match" if the ego rated both themselves and their alter identically. Note that the ego's information was collected in a *separate* survey from the ego-network survey (a demographic survey), preventing immediate ordering or cognitive consistency effects.

- *Open-Ended Activity Matching*: A count variable (ranging from 0 to 5) capturing the number of shared specific activity nominations. Egos were asked to freely list up to five leisure activities they enjoyed with the prompt: "Please list up to 5 specific activities that you enjoy doing in your free time." Subsequently, for each alter, the ego was asked to check boxes indicating which of these specific activities the alter also enjoyed, prompted with: "Which of your favorite activities does this person also enjoy?"

- *Cultural network opacity*: A count variable (ranging from 0 to 6) measuring the number of cultural domains for which the ego responded "Don't know" regarding their alter's preferences. For the closed-form domain questions regarding alters, respondents were provided a fourth response category: "Don't know." While survey researchers often instinctively treat "Don't know" responses as missing data to be imputed or dropped, we explicitly operationalize these responses as substantive data. In the context of ego-network surveys, a "Don't know" response is not random missingness; rather, it is a direct behavioral measure of relational ignorance (opacity). By retaining and counting these responses, we treat opacity as a distinct structural feature of the dyadic tie.

3.2.3 Additional Predictors

- *Subjective Closeness*: Measured with the prompt, "How close are you to this person?" Response categories were "Not close" (1), "Somewhat close" (2), and "Very close" (3). This is treated as a factor variable in the models. As Marsden and Campbell (1984) noted in their classic paper, this item is the most reliable indicator of tie strength (Granovetter, 1973).
- *Relationship Type*: Derived from the prompt "How are you connected to this person?" We include binary indicators for whether the tie is a `Friend` (vs. other types) or a `Roommate/Dormmate` (capturing on-campus structural propinquity).
- *Frequency of Interaction*: Measured with the prompt, "How often do you communicate with this person?" Responses were dichotomized into a `High Frequency` indicator (`Daily` or `Weekly`) versus the reference category (`Monthly` or `Less often`).
- *Tie Duration*: A continuous measure of how long the ego has known the alter (in years), asked via "How long have you known this person?" We include both the linear and squared terms (*Duration Squared*) to account for nonlinearities in the liability of newness.
- *Demographics*: We control for ego gender and alter gender (1 = Woman, 0 = Man), along with a multiplicative interaction term to capture same-gender vs. cross-gender dynamics. We also include a measure of *Race Homophily*, constructed by matching self-reported race for the ego and self-reported race for the alter. Ties are classified as `Homophilous` (same race), `Heterophilous` (different race), or `Unknown` (if either ego or alter race was missing).
- *Time Period*: Dummy variables for each wave transition interval (Wave 1 to 2, Wave 2 to 3, Wave 3 to 4, Wave 4 to 5, Wave 5 to 6, Wave 6 to 7, Wave 7 to 8) to allow the baseline hazard of tie persistence to vary across the entire panel span.

3.3 Analytical Strategy and Model Specification

Because our data tracks relationships over time and individuals can nominate multiple alters, our data structure consists of dyad-periods nested within egos. To properly analyze tie persistence, capture the high baseline volatility of personal networks, and account for right-censoring, we employ a discrete-time survival analysis framework (event history modeling). Since the outcome variable—whether a tie persists into the next wave—is binary, we estimate the hazard of tie survival (the probability of persistence given the tie survived to the previous wave) using mixed-effects logistic regression. This modeling strategy allows us to explicitly model the baseline hazard rate of tie decay across each wave transition, isolating the protective effects of our cultural variables against this background volatility.

To address the non-independence of observations arising from the clustered nature of the ego-network data (multiple ties per ego), we include random intercepts for egos. This accounts for unobserved, time-invariant individual heterogeneity in the baseline propensity to maintain ties.

The models take the following general form:

$$\text{logit}(P(Y_{ijt} = 1)) = \beta_0 + \beta_1 X_{ijt} + \beta_2 C_{ijt} + u_i \quad (1)$$

where Y_{ijt} is the indicator of tie persistence for ego i and alter j at time t, X_{ijt} represents the vector of our core theoretical predictors (cultural matching and cultural network opacity), C_{ijt} is a vector of dyadic and structural control variables (including time period indicators to allow the baseline hazard to vary non-parametrically over waves), and u_i is the ego-specific random intercept, assumed to be normally distributed with mean zero and variance estimated from the data.

All models are estimated via maximum likelihood using the `glmer` function from the `lme4` package in R. In models that interact subjective closeness with our cultural variables, we calculate and plot predicted probabilities to facilitate the interpretation of the non-linear interaction terms (Mize, 2019).

To verify the robustness of our core findings to modeling specification decisions, we conduct a supplementary analysis substituting our mixed-effects modeling framework for an ego fixed-effects framework using conditional logistic regression. This adjusts for all unobserved, time-invariant ego-level characteristics.[2]

4 Results

We present our findings in three stages. First, we examine the baseline main effects of cultural matching and opacity on tie persistence. Second, we test the Strength-Mediated Matching Hypothesis by interacting our matching variables with subjective closeness. Finally, we assess the robustness of our findings using ego fixed-effects models.

4.1 Main Effects: Cultural Matching and Cultural Network Opacity

Table 2 presents the odds ratios from a series of mixed-effects logistic regression models predicting tie survival. Across all models, the structural and dyadic control variables operate exactly as standard network theory would predict: ties that are subjectively closer ($OR = 2.45, p < 0.001$ for "Close" ties in Model 1) or more frequently interacted with ($OR = 2.14, p < 0.01$) are significantly more likely to survive.

Building on this structural baseline, Model 1 demonstrates that **closed-form cultural matching** significantly increases the odds of tie persistence. For each additional broad cultural domain in which an ego and alter share tastes, the tie is 9.1 % more likely to survive into the subsequent wave ($OR = 1.09, p < 0.01$), net of controls.

In Model 2, we introduce **open-ended activity matches**. Specific, shared behavioral practices provide additive protective value: each shared activity increases the odds of tie survival by 8.7 % ($OR = 1.09, p < 0.01$). When both are included, closed-form matching remains significant ($OR = 1.08, p < 0.05$), showing that specific activities provide protective value above and beyond broad categorical taste similarities.

[2]For a full summary of descriptive statistics, see Tables 3 and 4 in the Appendix.

Model 3 tests the independent main effect of **cultural network opacity**—measured as the number of domains in which the ego reported "Don't Know" regarding their alter's preferences. While opacity predicts a 4.9 % decrease in the odds of tie persistence per unknown domain ($OR = 0.95$), this main effect is not statistically significant at the conventional level ($p > 0.05$) when controlling for both closed and open-ended matches simultaneously. The attenuation of this coefficient in the pooled mixed-effects model is primarily due to the mechanical, between-person correlation between reporting "Don't Know" and having fewer known closed-form matches. However, as we will show in the interaction models and the fixed-effects robustness check below, opacity remains a profound and significant liability within specific relational contexts.

To contextualize these coefficients, Figure 1 plots the average marginal effect of each core theoretical variable on the probability of tie persistence. Across the board, cultural matching serves as a protective mechanism against tie decay (positive estimates), whereas cultural network opacity acts as a liability (negative estimates).

4.2 The Moderating Role of Subjective Closeness

While the main effects demonstrate that cultural alignment generally stabilizes ties, we next assess whether this "social glue" is uniformly necessary for all relationships. We estimated auxiliary interaction models (not shown in tables, but visualized via predicted probabilities) to test whether the protective effects of cultural matching and the detrimental effects of cultural network opacity are moderated by baseline subjective closeness.

Because interaction terms in logistic models can be opaque due to the non-linear link function, we visualize this critical interaction to make the core finding instantly digestible. Figure 2 plots the average marginal effects of our three cultural variables across ties rated as "Not Close," "Somewhat Close," and "Close."

The results reveal a nuanced asymmetry: the moderating role of subjective closeness depends heavily on the type of cultural measure. We find a strong interaction for both closed-form cultural matches and cultural network opacity, but not much of an interaction for open-ended matches. For broad, closed-form matches and network opacity, the predictive slopes (marginal effects) are steepest and most significant for *strong* ties (those rated "Close"). For these essential relationships, an increase in shared cultural tastes dramatically increases the predicted probability of tie persistence. Correspondingly, cultural network opacity exerts its most severe, statistically significant penalty precisely on these strong ties: lacking mutual cultural knowledge heavily accelerates their decay. Conversely, for weak ties (those rated "Not Close"), the marginal effects for both closed-form matching and opacity are relatively flat, indicating an intrinsic fragility to decay regardless of cultural alignment or opacity in these broad domains.

This interaction suggests that shared cultural tastes and mutual cultural knowledge act as the substantive "core" of deep, institutionalized friendships, requiring ongoing cultural alignment to be maintained. Weak ties, on the other hand, are fleeting and superficial; their baseline hazard of decay is so high that cultural matching does little to save them. In contrast, the protective effect of specific, open-ended activity matches operates more uniformly across relationships of all strengths, providing a consistent anchor for tie persistence whether the tie is new or well-established.

4.3 Robustness Check: Ego Fixed-Effects

To confirm that our findings are not artifacts of our primary mixed-effects specification, we estimated an **Ego Fixed-Effects model** via conditional logistic regression (shown in Table 1). By stratifying the likelihood function by ego, this approach leverages only within-ego variance—comparing the alters a single ego retains versus the alters that same ego drops. This stringent test perfectly isolates dyadic tie characteristics from unobserved individual baseline differences (e.g., an ego's baseline sociability, introversion, or general propensity to use the "Don't Know" survey option).

As shown in the table, the fixed-effects models yield results that are remarkably consistent with our random-effects main models. Closed-form matches ($OR = 1.07, p < 0.10$) and open-ended matches ($OR = 1.11, p < 0.001$) remain positive predictors of tie persistence, with odds ratios that are nearly identical in magnitude to the baseline models. Crucially, in this strict within-ego specification, **cultural network opacity** emerges as a highly significant negative predictor of tie persistence ($OR = 0.91, p < 0.05$), reducing the odds of tie survival by 9.1 % per unknown domain. This confirms that while opacity may correlate with overall match rates across individuals, *within* an individual's personal network, a lack of cultural knowledge about a specific alter is a powerful, independent signal of tie dissolution. The structural controls—such as being "Close" ($OR = 2.82, p < 0.001$) or having weekly/daily contact ($OR = 1.91, p < 0.05$)—also maintain their robust associations.

5 Discussion and Conclusion

5.1 Summary of Key Findings

In this paper, we sought to address a central issue in the study of culture and networks: how do cultural tastes shape the evolutionary dynamics of personal networks? Using longitudinal ego-network data from the *NetSense* study, we considered the empirical implications of a cultural matching model of tie persistence. Our results provide strong evidence that shared cultural tastes are a vital mechanism for stabilizing social ties. Specifically, both broad, closed-form cultural matches and specific, open-ended activity matches independently reduce the hazard of tie dissolution. Furthermore, we identified cultural network opacity—the lack of mutual cultural knowledge—as a powerful, independent liability that accelerates tie decay. Crucially, we found that the protective effect of closed-form matching and the detrimental effect of cultural network opacity are most pronounced among strong ties. For these subjectively close relationships, cultural alignment acts as a vital core for tie survival, whereas weak ties appear so fleeting that their baseline hazard of decay is high regardless of matching in these domains. In contrast, the protective effects of specific, open-ended matches remain relatively consistent across both weak and strong ties.

Table 1: Robustness Models (Ego Fixed-Effects)

	Ego FE (clogit)
Closed-Form Matches	1.065+
	(0.037)
Open-Ended Matches	1.112***
	(0.035)
Network Opacity	0.909*
	(0.034)
Subjective Closeness: Somewhat	1.575***
	(0.204)
Subjective Closeness: Close	2.815***
	(0.401)
Roommate/Dormmate	0.753+
	(0.122)
Frequency: Weekly/Daily	1.906*
	(0.552)
Race Homophily: Same Race	0.948
	(0.114)
Tie Duration (Years)	0.983
	(0.042)
Num.Obs.	6054
AIC	5030.1
BIC	5144.2

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Note: Controls for ego and alter gender, non-linear tie duration, and period fixed effects.

5.2 Theoretical and Substantive Interpretations

These findings carry significant theoretical implications for how we understand the relationship between tastes and networks, particularly regarding the role of tie strength. By adjudicating between our competing hypotheses, our results heavily favor the **Strong-Tie Matching Hypothesis** over the Weak-Tie Scaffolding Hypothesis. Aligning with previous research (Edelmann and Vaisey, 2014; Vaisey and Lizardo, 2010), our longitudinal survival framework reveals that deep, institutionalized friendships require ongoing cultural resonance to sustain intimacy. It is the *strong* ties that are highly sensitive to cultural matching and opacity. Weak ties, suffering from a high liability of newness, rapidly decay with or without shared tastes.

More broadly, these results challenge the assumption of the pure network transmission perspective, which treats cultural tastes merely as fluid by-products of existing social ties (Mark, 2003). Instead, our findings align more closely with the cultural capital perspective (Bourdieu, 1984), suggesting that tastes function as durable, portable resources that individuals actively use to select, maintain, and discard social relationships.

Returning to the puzzle posed in the introduction—how do we account for durable cultural profiles in the face of volatile networks (Bourdieu, 1984; Holt, 1998)—our results help to close the loop. If persons use relatively stable cultural dispositions to selectively maintain their networks, it suggests that cultural capital is not merely a passive status marker or a "thin" effluvium of network position. Rather, it is an active navigational tool for managing social environments. People churn through weak ties at a high baseline rate, but they rely on their durable cultural compass to actively curate and protect the core relationships that matter.

Furthermore, our findings speak directly to the classic distinction between choice homophily and induced homophily (McPherson et al., 2001). If cultural matching actively prevents tie decay and opacity accelerates it, then a substantial portion of the observed homophily in established social networks is likely due to selective retention (choice homophily) rather than just structural constraint or network-induced adaptation. By retaining culturally matched ties and shedding incompatible ones over time, networks become homophilous through an active evolutionary sorting process.

In this paper, we conceptualize cultural matching as the essential structural core of close relationships. While early network theories emphasized structural constraints or multiplexity (Burt, 2000), our findings suggest that shared tastes may be what gives close relationships their staying power. In the absence of mutual cultural knowledge (high opacity), even subjectively close ties face a higher hazard of dissolution. This developmental view of networks helps reconcile the volatility of personal networks with the relative stability of individual cultural profiles.

Beyond sociological theory, these findings carry practical implications for institutions aiming to foster social cohesion. For university administrators and community organizers seeking to reduce social isolation, our results underscore that mere proximity or shared structural affiliations are insufficient to sustain durable relationships. Facilitating environments where individuals can discover and activate shared cultural interests—from niche clubs to diverse recreational programs—may be essential for transforming fleeting initial encounters into resilient, long-term support networks.

5.3 Limitations and Future Research

Despite these promising findings, several limitations point toward fruitful avenues for future research. First, our empirical focus on a sample of college students at a single university limits the generalizability of our claims. The college transition is a period of heightened network flux and identity formation. Future research should examine whether cultural matching operates similarly as a stabilizing force in adult populations, workplace settings, or among older adults (Cornwell et al., 2008) where network dynamics and the salience of different cultural domains may differ.

Second, the current measurement approach treats most cultural matches as theoretically equivalent. Future research could further disaggregate these domains to test whether high-status cultural matches (e.g., classical music, literature) confer different relational advantages compared to popular or low-status domains (DiMaggio, 1987). Additionally, researchers could explore whether shared dislikes function as a separate dimension of boundary-drawing.

Finally, while our discrete-time survival framework effectively models tie decay, future studies would benefit from deploying stochastic actor-oriented models (SAOMs) to simultaneously model tie creation, tie dissolution, and individual taste change (Hachen et al., 2022). Such an approach could fully disentangle the reciprocal processes of network-based cultural influence and culture-based social selection.

Figures

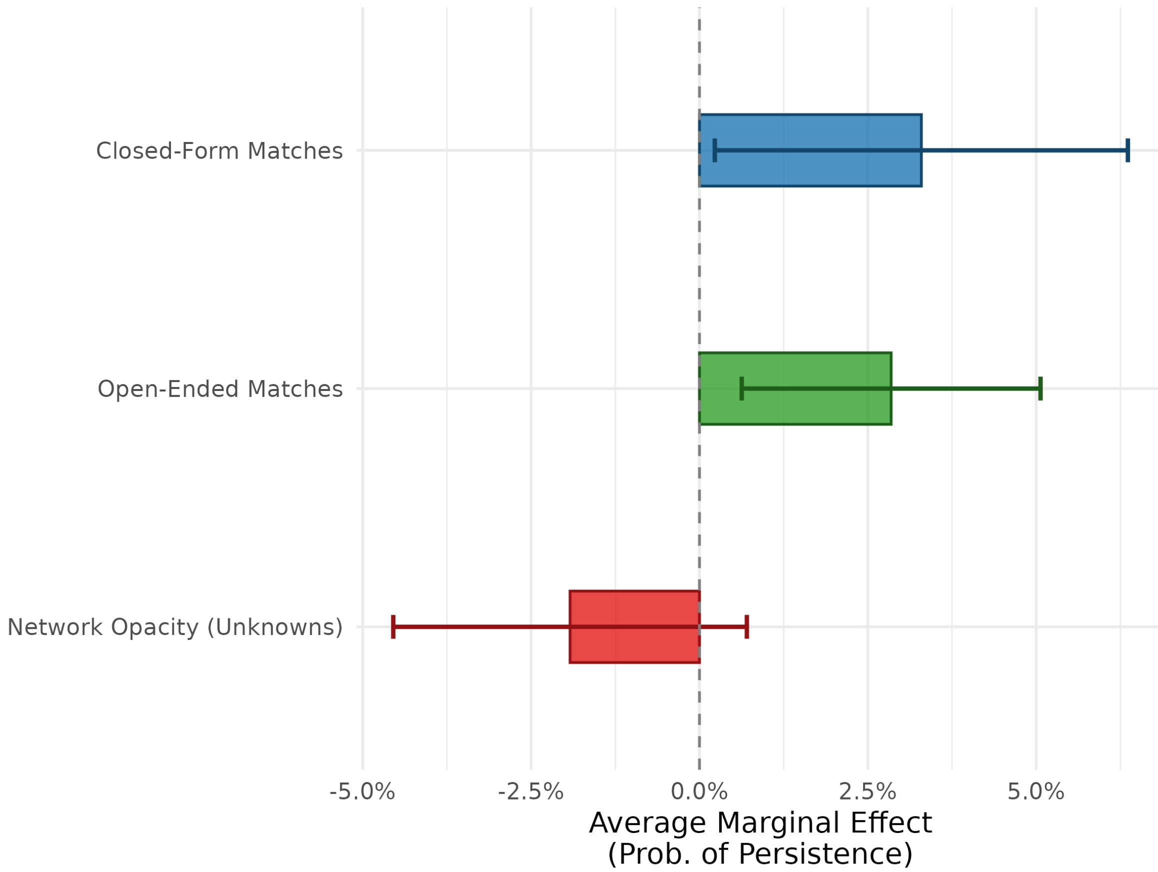

Figure 1: Predicted Probability of Tie Persistence by Cultural Matching and Cultural Network Opacity.

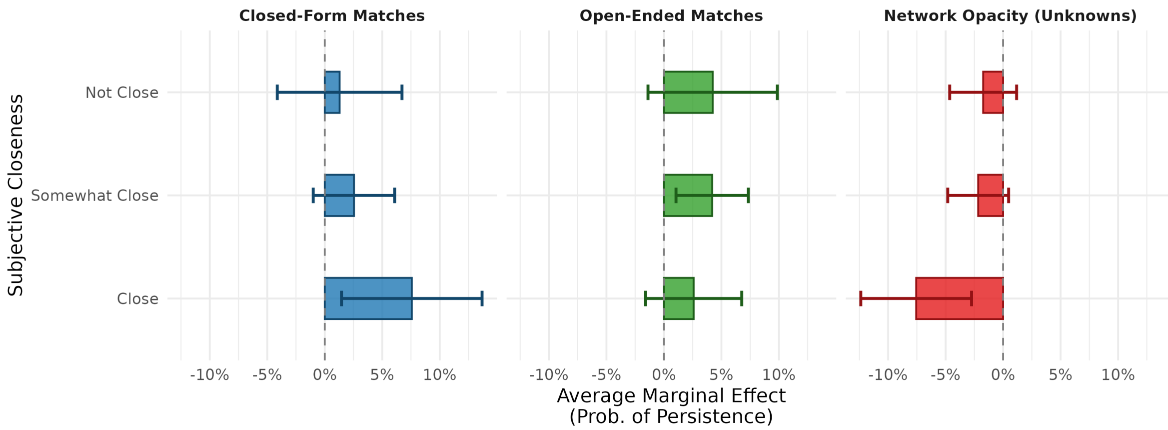

Figure 2: Predicted Probability of Tie Persistence by Cultural Matching, Cultural Network Opacity, and Subjective Closeness.

Tables

Table 2: Odds Ratios for Tie Persistence (Main Effects Models)

	Model 1: Closed	Model 2: + Open	Model 3: + Opacity
Closed-Form Matches	1.091**	1.076*	1.063+
	(0.035)	(0.035)	(0.036)
Open-Ended Matches		1.087**	1.084**
		(0.034)	(0.034)
Network Opacity			0.951
			(0.036)
Subjective Closeness: Somewhat	1.419**	1.359*	1.332*
	(0.189)	(0.183)	(0.180)
Subjective Closeness: Close	2.454***	2.285***	2.215***
	(0.347)	(0.329)	(0.324)
Roommate/Dormmate	0.794	0.802	0.800
	(0.128)	(0.129)	(0.129)
Frequency: Weekly/Daily	2.136**	2.146**	2.209**
	(0.523)	(0.526)	(0.545)
Race Homophily: Same Race	0.955	0.948	0.939
	(0.121)	(0.121)	(0.120)
Tie Duration (Years)	0.991	0.988	0.984
	(0.044)	(0.043)	(0.043)
Num.Obs.	5497	5497	5497
AIC	5256.3	5250.7	5251.0
BIC	5368.7	5369.7	5376.6

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001
Note: All models control for ego and alter gender, non-linear tie duration, and baseline hazard variations across all 7 wave transitions (coefficients omitted for space). Random intercepts for egos are included.

Table 3: Descriptive Statistics (Continuous Variables)

	Mean	SD	Min	Max
Closed-Form Matches	1.56	1.50	0.00	6.00
Open-Ended Matches	2.48	1.35	0.00	5.00
Cultural Network Opacity	0.52	1.23	0.00	6.00
Tie Duration (Years)	3.98	6.00	0.00	21.16

A Appendix: Descriptive Statistics

Descriptive statistics for all continuous and categorical variables used in the discrete-time survival analysis models.

Table 4: Descriptive Statistics (Categorical Variables)

		N	Percent
Tie Persisted	0	4311	70.18
	1	1832	29.82
Ego Gender	Men	3007	48.95
	Women	2569	41.82
Alter Gender	Men	3131	50.97
	Women	2991	48.69
Subjective Closeness	Not Close	681	11.09
	Somewhat Close	2890	47.05
	Close	2504	40.76
Interaction Frequency	Daily	184	3.00
	Less often	5933	96.58
	Weekly	26	0.42
Race Homophily	Heterophilous	624	10.16
	Homophilous	1786	29.07
	Unknown	3733	60.77

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